

ASCII MASTER FLOW – PANEL 1/12: Nuclear-process firewall

FISSION -> splits heavy nuclei and powers current reactors

FUSION -> combines light nuclei in experimental systems

COMMERCIAL STATUS -> fission operating, fusion not deployed

SHARED WORD NUCLEAR -> does not erase process differences

RULE -> Topic 05 cannot substitute for Topic 04

ASCII MASTER FLOW – PANEL 2/12: D-T energy map

DEUTERIUM + TRITIUM -> fusion reaction

HELIUM-4 / ALPHA -> charged product retained for self-heating

NEUTRON -> carries most energy out of plasma

BLANKET -> captures heat and supports fuel functions

RULE -> particle role controls engineering consequence

ASCII MASTER FLOW – PANEL 3/12: Fusion-condition triangle

TEMPERATURE -> raises probability of overcoming repulsion

DENSITY -> supplies reacting particles

CONFINEMENT TIME -> retains energy long enough

LAWSON TRIPLE PRODUCT -> all three must combine

TRAP -> a temperature record alone proves little

ASCII MASTER FLOW – PANEL 4/12: Confinement comparison

MAGNETIC -> lower density and longer confinement

TOKAMAK -> torus with strong plasma current

STELLARATOR -> externally shaped steady-field geometry

INERTIAL -> dense target and extremely short confinement

RULE -> approach and device name are not power status

ASCII MASTER FLOW – PANEL 5/12: Breakeven ladder

PLASMA Q -> fusion power divided by injected heating

Q ABOVE ONE -> plasma-level scientific threshold

ENGINEERING Q -> includes recirculating plant loads

NET ELECTRICITY -> includes thermal conversion and all consumption

COMMERCIALITY -> adds availability, cost and licensing

ASCII MASTER FLOW – PANEL 6/12: Neutron-to-material chain

14.1 MeV NEUTRON -> escapes magnetic confinement

BLANKET -> receives energy and may breed tritium

STRUCTURE -> displacement damage and transmutation

ACTIVATION -> radioactive material-management issue

LIFETIME -> separate engineering feasibility constraint

ASCII MASTER FLOW – PANEL 7/12: Tritium fuel loop

D-T PLASMA -> consumes tritium

FUSION NEUTRON -> enters lithium-bearing blanket

LITHIUM REACTION -> produces new tritium

EXTRACTION / PROCESSING -> returns usable fuel

RULE -> a test blanket is not a closed commercial cycle

ASCII MASTER FLOW – PANEL 8/12: Plasma exhaust path

CORE PLASMA -> creates heat and helium ash

EDGE TRANSPORT -> moves particles and heat outward

DIVERTOR -> exhausts impurities and helium

PLASMA-FACING MATERIAL -> receives extreme heat load

RULE -> confinement success does not solve exhaust lifetime

ASCII MASTER FLOW – PANEL 9/12: ITER mission card

UNDER CONSTRUCTION -> experimental international tokamak

BURNING-PLASMA STUDY -> core scientific purpose

Q=10 DESIGN -> 500 MW fusion from 50 MW plasma heating

NO TURBINE -> no conversion to grid electricity

TEST TECHNOLOGIES -> bridge to later machines

ASCII MASTER FLOW – PANEL 10/12: India-in-ITER map

INDIA -> one of seven ITER members

IN-KIND PACKAGES -> specified hardware and systems

IPR -> DAE-aided plasma-research institute

ITER-INDIA -> domestic procurement and delivery agency in IPR

RULE -> membership, institute, agency and package differ

MUST REMEMBER: Fusion joins light nuclei in a high-temperature plasma; confinement,...

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ASCII MASTER FLOW – PANEL 11/12: Domestic research devices

ADITYA-U -> experimental tokamak and plasma campaigns

SST-1 -> superconducting long-pulse research platform

RESEARCH OUTPUT -> diagnostics, control and operating knowledge

NOT OUTPUT -> commercial fusion electricity

RULE -> experimental operation is not power generation

CLOSE DISTINCTION: Plasma Q is not engineering breakeven, tokamak is not stellarator,...

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ASCII MASTER FLOW – PANEL 12/12: Fusion-readiness answer spine

DEFINE -> D-T fusion plasma confinement Lawson and Q
TRACE -> alpha heating neutron blanket tritium and materials
SEPARATE -> plasma gain engineering gain and net electricity
MAP -> ITER India IPR ITER-India ADITYA-U SST-1 and DEMO
VERIFY -> dated schedule cost contribution and milestone evidence
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State quantity, device, pulse/energy...
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